

Mahault Albarracin ~ Active Inference Insights 002 ~ Epistemic Communities

Insights | Active Inference Insights 002 ~ Epistemic Communities, Social Scripts, AI | 1:47:18

Hello everybody and welcome back to the Active Inference Insights podcast brought to you by the Active Inference Institute. I'm your host, Darius Parvezi-Wayne, and today I have the pleasure of chatting to Mau Alboracen. Mau is a Cognitive Computing PhD candidate at the University of Quebec in Montreal, as well as the Director of Product Innovations and Research at Versys, a company which leads the way in applying the Active Inference Framework to AI.

With expertise in computational neuroscience, sexology, consciousness, and much more, Mau's work is broad, innovative, and deeply exciting, which makes me rather excited for this conversation. Mau, welcome to the show. Thank you so much for joining us. It's, as I said, it's an absolute pleasure. Well, thanks for having me. So, yeah, as I sort of mentioned in the introduction, your work covers a very broad range of topics and areas of research, but as you know, this podcast is exploring the themes of Active Inference and the Free Energy Principle. So, I want to mainly focus on that, even though we can see about how these different themes tie in with that overarching framework. If one goes on your Google Scholar and sees your work on Active Inference, again, they'll see it's diverse. Where I want to start is the paper that you wrote on epistemic communities. I guess the first question our audience might have, just as a reminder for those listening, this podcast is very much meant as a gentle introduction to Active Inference, as gentle as an introduction to Active Inference can be. So, just for our audience, my first question is very simple, and it's just, what is an epistemic community? So, it has many definitions. It's a loosely used term in a variety of fields, but the way we've used it is essentially a community of people that share very specific types of beliefs that lead them to sample reality in the same way. So, if you think about, like, your norms, they usually are tied to a specific way of thinking. Like, for instance, the fact of going to work. The fact of going to work nine to five is tied to an ideology that values a certain type of productivity, consumerism, etc., and a certain type of relationships, where we are together in the morning, together in the evening, and we are with different people during the day, etc., etc., etc. So, your life becomes shaped by these ideas that give rise to norms. Cool. And let's sort of tie in the Active Inference Framework then to this idea that we have a community which is tied together through sampling similar norms, similar sensory observations. Perhaps you could start just by giving us a brief or, yeah, relatively accessible notion of what Active Inference is.

I mean, Carl did a very good job last week at explicating this, maybe just as a reminder to people. And then how that framework and the truths that the free energy principle discloses can be applied to the formation of an epistemic community.

Yeah. So, Active Inference as a principle is rather simple. It's really just the idea that we try to minimize how much we are likely to dissolve. We don't want to die. So, in order to do so, we're going to try to map the world and understand what might be coming such that we can avoid moments where we might be dissolving, or moments that are dangerous for us. More

surprising. We try to create evidence of ourselves. We try to predict ourselves existing in the future.

And to do this, we make a prediction, act in the world, try to change something, see if it changes the way we thought it was going to change, and then update our model of the world. So you've got this sensory active loop where the agent is embedded in the world rather than just outside of it, mapping the world.

So, it's a pretty crucial idea that the agent necessarily has a perspective which is defined by how they are separate and yet embedded in the world. They have a boundary. We generally refer to that as a mark of blanket. This boundary is both defining and defined by the outside world and the agent themselves.

So, if you understand that the agent necessarily has to map the world in order to predict what's likely to happen, but that the agent is only capable of dealing with their perspective and how much they can compute from their perspective, you understand the agent is fundamentally limited.

That's where they are known to make this skill to make this skill to make this skill to make this skill to make
this skill to make this skill to make this skill to make this skill to make this skill to make this skill to make this
skill to make this skill to make this skill to make this skill to make this skill to make this skill to make this skill
to make this skill to make this skill to make this skill to make this skill to make this skill to make this skill to
make this skill to make this skill to make this skill to make this skill to make this skill to make this skill to make
this skill to make this skill to make this skill to make this skill to make this skill to make this skill to make this
skill to make this skill to make this skill to make this skill to make this skill to make this skill to make this skill
to make this skill to make this skill to make this skill to make this skill to make this skill to make this skill to
make this skill to make this skill to make this skill to make this skill to make this skill to make this skill to

if they've ever found themselves down the rabbit hole of an echo chamber, whether that's on Twitter or Instagram or wherever.

I guess a question that comes to my mind here is that maybe a heuristic idea would be that truth exists as some kind of prior within a generative model that an agent like us has.

It's important that we tie ourselves in some way to veracity, because if not, then we're going to be constantly surprised every time something false happens.

Or that false true distinction is a bit blurry, but you see where I'm going.

So where I'm going with this is that a lot of echo chambers are reinforcing their model of the world,

but it doesn't necessarily need to subscribe to any notion of external truth.

To what degree does that ever butt up against a kind of inherent prior, which is that eventually the falsity of your claims is going to come home?

Or can the echo chamber, the entrenchment, just keep going, keep digging as far down as possible?

Yeah, so this is a super interesting idea.

And we first have to define what we mean by truth and what level of truth.

What do we mean by truth?

Do we mean something that I can observe?

Do we mean something that is verifiable by other agents?

And then what level of truth do we mean?

The truth of the event, the social truth of the meaning of the event,
the fact that to some extent, no matter what happens,
this element will be considered true.

Do you see what I mean?

Like there's so many ways in which this can break down.

And what this leads us to is, so this is a theory.

This is something I'm trying to develop.

Obviously, it's not verified yet.

So it's still something to take with a few grain of salt.

But it's a theory that I think naturally falls from what we've done.

Depending on the level of truth you're talking about and your definition of truth,
the fact that something, say, happened or not can be irrelevant.

What matters about truth is whether it's actionable to some extent.

So let's take an example.

If I decide I can fly, but I never act on it,
it has no impact on my life.

My sensory active loop isn't going to change.

My model is not going to change.

So I can carry this belief without ever having any consequence at that level.

I'm not going to die.

But now I can tell other people that I can fly.

And if none of us act on it, but we all squarely believe we can fly.

And it serves a social purpose of identifying each other.

We know who we are.

And then we congregate to each other.

We create a reality together.

We create purposes.

What happens then?

Where does the buck fall?

Where does that belief become a belief that potentially harms us?

I think that's a true question.

So think of things that are more contentious, like does God exist?

Technically, there is no way to prove it.

Literally.

Part of the definition is that believing God has to be faith-based.

Or at least, you know, for Christian, Judeo-Christian beliefs.

It has to be faith-based.

Which means by definition, you shouldn't need evidence.

You should believe without evidence.

You should take a leap of faith.

Okay.

Then by virtue of that, we can't verify it.

We can't change it.

Your belief is kind of impervious.

So then what are the outcomes of that belief?

How does it allow a community to survive?

How does it allow people to survive?

And there's a really interesting phenomenon that happens there.

Some people will start manufacturing evidence of God.

They will start thinking, okay, well, yes, it's true.

I don't need evidence.

But I can read certain kinds of things as evidence.

And they will select the kinds of things that count as evidence, i.e.

things that they can't ascribe a clear cause to.

So, for instance, they won't ask God to feed their dog.

Because if they did, and the dog doesn't get fed, the dog dies.

And now they're like, well, God didn't do it.

There's a clear proof here it didn't happen.

But something more nebulous, like in the future, I will get a job.

I mean, you know, like it's highly likely you're gonna.

But can you really ascribe the cause of that?

Like you might have sent some resumes, but there's no guarantee anyone will pick up your resume.

Like it's practically luck at this point.

So if it's luck, it's God.

Because luck is not something we can ascribe a clear cause to.

So then you get these communities that are functional because that belief never really comes into play in their actions.

And so you can have weird types of layered beliefs for certain people.

Like you could have a flat earther be your manager.

It probably won't change much.

It's not compatible with your beliefs.

But in terms of the layer that you share, the functional layer that you share, it doesn't come into play.

But if I believe I can fly and therefore you can fly and I tell you, go, go, do it now.

Fly.

Jump off the building.

Now we're gonna have a problem.

And therefore that kind of belief is less likely to be carried because functionally it's likely to disappear.

So in terms of self-organization, it just is likely to disappear.

Right.

Yeah, that's interesting.

I guess it's absolutely right to unpick these different layers.

And the fact that I guess one way of looking at it is that truth can have this kind of instrumentalist or pragmatic aspect to it,

but also this something that, you know, philosophers of language would think about a little bit more in terms of conditional statements and whether something's actually true in the world.

I wonder whether, therefore, if we look at the active inference framework and we try to apply active inference to this,

a question that comes up in my head is does for an agent to consider their belief to genuinely be true?

Because I think what we can say here is if I had the belief that I could fly, perhaps I occupy that truth at an instrumentalist level because I'm surrounded by other people who believe that they can fly and it actually gives me a sense of purpose.

And so I almost delude myself into thinking that, but really a push came to shove and someone had me at the top of a cliff edge.

I probably wouldn't do it because I don't think there's actually a state of the world out there which says Darius can fly.

Does the, all of that to say that does, does the active inference framework require for an agent to believe that there is genuinely solid epistemic grounds to believe that something is ontologically the case

so that I can actually fly that if I took off that I would need to be able to sample or have sampled evidence for that?

Or is there a mechanism that might be peripheral to that which could give me that trust in a sufficient trust in my model that I maybe I would jump off the cliff?

So I think we have to look at how life is realized and how you plan yourself in the future, right?

So let's assume you really think you can fly and your life just doesn't lend itself to it.

Like, that's not what you do.

You take the car, you, you know, you hang around, you have coffee, you work, whatever.

It's just, it just never comes into play.

You could hold that belief with high certainty because it never gets tested and there's never a risk.

Now, if you were to start having a real risk, like, it actually, like, that state is not highly unlikely.

It's like, it's over here.

And you now have to start testing your own predictions of yourself into the world.

And you're going to start seeing spikes of free energy like this.

Like, you've never seen anyone fly.

You sure you can, but you've never seen anyone fly.

And the risk associated with not flying is really, really, really high.

I think that's when you start seeing the armor break.

Like, I think the difference becomes between someone who has lost touch with actual reality.

Like, they genuinely believe they can fly and it becomes an actual thing they need to do now.

They probably will do it.

But they will also probably die.

Which means that these kinds of beliefs, which we generally consider to be psychotic, right?

Like, it's, it's, that's the definition.

It's a loss of touch with reality, no matter what that means.

And so when we start couching these ideas in this way, we start depathologizing a little bit.

We're like, well, it's not, they're not, they're not sick per se.

There's nothing intrinsically wrong with them.

But their beliefs are no longer aligned with the ecosystem within which we exist.

If they were on, say, the moon, for a minute, maybe, you know, they could fly.

Like, it wouldn't be an impossible belief.

It's just not a belief here that works.

So I think that's why most of the people who hold beliefs that are not related to what we would consider something that happened.

Like, it's not verifiable.

It's not, it's not something that we can assure, ensure happened or can happen.

But, you know, generally, these beliefs don't relate to things that are possibly harmful to them.

Like, for instance, believing in Trump, believe in what Trump says.

It's easy to debunk what he says.

Like, a lot of people have done it.

In fact, it's actually, it takes more time to debunk everything he says than just to listen to him.

But it has no impact on your life if you do believe him.

Like, pragmatically, it won't hurt you.

So it's a very useful tool to lie about things that are completely unverifiable to you.

Therefore, why not believe them?

Yeah, I think what you spoke about in terms of depathologizing is very interesting.

When I was speaking to Carl, for those who are listening, they might recall we sort of spoke about the fact that the complete class theorem states that actually any behavior can be cast as Bayes' optimal under some prior beliefs.

So I think that's a nice, it leads itself, it lends itself to a humane approach to psychopathology or confirmation bias, less, you know, more mundane instances of circular reasoning or convincing yourself that something is true when it's clearly not.

Going back to your paper, I thought an interesting thing.

So for those who don't know, Mao and Maxwell and others sort of created a computational model in addition to the theorizing.

Right. So there is the theory and then you would put it in silico and you endowed your silico agents with an a priori hypothesis that or preference that they would be more likely to sample agents who are similar to them.

Which is, in a sense, the kind of prior preference which underwrites confirmation bias.

My question here is, in that are you stipulating that we too have those prior preferences?

Because on one hand, yes, I mean, that manifests in the behavior and it makes sense in terms of confirming your own model.

At the same time, humans can also be incredibly flexible creatures who do change their model on the basis of evidence.

So firstly, is the claim that that is an actual feature of human cognition and our generative model?

And two, if so, would you couch that in terms of natural selection and evolutionary pressures?

Or is that is even more fundamental to self-organization to sample information from those who are more similar to you?

Right. So confirmation bias was a parameter. It was something we tested with and without or on a continuum.

And we saw what the effects of it were relative to the formation of echo chamber or to the change in beliefs.

And then we saw that it had a magnifying effect, but it wasn't as simple as just you turn it off and it's gone and you turn it on and it's there.

Right. There was still the possibility to form communities without the notion of confirmation bias.

It just wasn't as strong. And it also depended on the network dynamics.

Different networks propagate information differently.

So we also had the notion of habit formation.

So if you have the need to have habits, basically you're going to slowly start sampling the same thing over and over and over again.

So there were a lot of parameters that we used and we found that confirmation bias basically biased the agent's likelihood model to acquire more evidence about their environment if the information confirms their pre-existing beliefs.

It was that simple, basically. So basically it just made it so that they would still sample people around them, but they were more likely to resample somebody that said something that agreed with their model, which is extremely compatible with active inference.

Right. You were trying to self evidence. So you're going to try to sample things that seem to agree with your model.

So confirmation bias was just a formalization of that specific parameter.

What we see is.

It's actually adaptive for you to be with people like you.

When you are with people like you, not only are you likely to understand people better and therefore share information more efficiently,

you're also likely to be in an environment that is beneficial for you, not just because you craft the environment around you,

but because if you find yourself with people like you, you're probably finding yourself in an environment that

benefits those people.

So, you know, it's just adaptive.

Now, it's adaptive to an extent.

If you're only with people with exactly your same model, then you're also very likely to have the same issues and to be vulnerable to the same kind of volatility.

It's like monoculture.

If you have one type of banana, if there's one disease banana, they're all going to get it and you no longer have bananas.

It's the same idea.

You want to continuously grow and change and gather more error and increase your fitness landscape such that you never get to a point where there's an error that none of you saw coming

and it just wipes everything out like a black swan event.

So there's like a nice middle ground.

It is.

It's like a super stimulus.

If you find yourself in an echo chamber, it feels really, really good.

For a time.

But there is a part of your model, the part that wants exploration, that eventually understands that you're stagnating.

And something needs to change.

Something needs to happen.

So it's not because we try to confirm our model that we also just want the model to be stationary.

We want to minimize free energy and we want to know that we are minimizing free energy.

Right.

It's layered and metacognitive.

So in that sense, you want to see yourself going down.

So if you're in a plateau, you're no longer going down.

Right.

So that's why I think to some extent, even when you have a group that seems to agree, that seems to have the same idea, it'll eventually start disagreeing inside.

Like there will be people that'll push, that'll keep polarizing such that now there's a little bit of a difference.

Like you're still, some people are like, well, no, okay, that's too far.

I don't think all of the world is flat.

Maybe, you know, a little bit, but not all of it.

Or I don't think the entire universe is just a planet.

That's silly.

Do you see what I mean?

Like it's the same belief.

It's the same kind of moving away from the trajectory that the others are going.

But you can't always constantly absolutely agree or you cease to learn information.

So you'll start again, sampling a little bit outside.

It's just that if you're stuck in an echo chamber and the echo chamber has, you know, drifted, drifted, drifted, drifted, drifted so far away from anybody you could ever talk to such that there is no reconciliation point here.

You're just going to have like a straight up no.

There's nothing to hold on to for you to be like, well, maybe my model is a little similar, but not quite the same.

So I could, you know, adapt a little bit.

But you're there's this this this ridge is too far.

It's too it's too far gone.

So it is adaptive for you to be with people like you.

But if it's too like you, then you're stagnating and eventually you're likely to potentially disappear.

Amazing.

Yeah.

Loads of things come to mind from that answer.

As you said, again, for those listening, as you said, just important to point out, as you said, to reduce free energy over time.

So-called expected free energy in the actual mathematical formula.

There is this balance of epistemic and pragmatic actions.

So we can't we do have to explore the world and we do have to sample in conditions of uncertainty.

So if people are curious, this this sort of evades the so-called dark room problem, which was a kind of an initial contention to the free energy principle, which people can read about.

And this over 10 years old now, these arguments.

I guess listening to this, something that comes to my mind is.

And then, well, something that comes to my mind and maybe those listening is why does one end up in these states anyway?

So maybe I'm listening to this and I might I'm not, but I might have been in some hypothetical world of flat earth and I might start panicking saying, well, what does that actually say about me?

Fundamentally, that I was amenable to those claims whatsoever.

Is there in some computational formulation or just some conceptual formulation an idea about why some people tend more than others to be wrapped up in conspiracy theory or group think this monolithic way of cognizing rather than cognitive flexibility?

And again, I don't mean to put these two on sort of binary polarities as if one is amazing and one's terrible.

As you say, we do spend time with people we agree with, all of us, as we should or else we'd have no friends.

But is there a reason perhaps why some people end up really going down that rabbit hole and suddenly, you know, and it's it's bizarre for those on the outside as to how they've got to that point?

Right. So let's always remember that anything we're going to say is a conjecture because social systems are extremely complex and it's a little bit like chaos theory, like ascribing one or two causes to a very complex phenomenon is always wrong.

We can have a model and it's an approximation, but we can't make too many blanket statements.

So that's how I'm going to caveat what I'm about to say next.

Second thing I'm about to say is another caveat is everyone does group think.

Everyone does it like it's just that some group think is considered.

And that's what I meant by the truth.

Yeah.

In the end, can you reproduce any of the experiments that come out from the scientific field?

Can you like reproduce a proof and you run it yourself?

Do you understand everything that comes out?

The truth is, no, you have to rely on others for an epistemic chain.

Like, is this a justifiable belief?

Yes, maybe.

I think, you know, that person looks really confident and other people seem confident in them.

So I'm, you know, follow that.

And sometimes what their model allows you to do is predictions about the world and those predictions happen to come true.

So you, through abductive inference, believe it must be true.

But it doesn't mean that.

In fact, it could be completely unrelated.

Maybe this is just like, you know, something that happens anyway.

If everybody told you, you've got to jump and the sun won't come up.

So we've jumped and the sun kept coming up.

And you're like, all right, I'll jump.

And you're like, you're right.

The sun didn't come up.

Wow, that works.

So, like, it could be completely wrong.

But you still, you know, that's your best explanation so far.

It allowed you a prediction.

Your prediction worked.

Okay, well, there you go.

There's some evidence for that.

In our case, it's more along the lines of our beliefs are the right beliefs.

And our ideology is the right ideology because it leads to XYZ kind of KPI we've arbitrarily defined as the KPIs that correspond to what allows us to minimize free energy.

But it's highly contentious.

Because oftentimes the KPIs we define are the KPIs that allow us to stay in the state that we are.

And so it's heavily slanted towards people in power, right?

Okay, propaganda, media, etc.

All of these tools are built to get you to believe that you are correct and you are doing the right thing.

And that right thing is to give more money to rich people.

So, you know, let's keep that in perspective for the time being.

Why do some people fall down rabbit holes?

It's a very known thing that there are key funnels for certain kinds of ideologies.

Because certain kinds of ideologies have a dynamic.

They very quickly allow you to feel better about yourself.

Feel better about yourself or feel more in control of the world.

Like you now have an answer.

You can now understand what's going on.

Your model feels like it reduced free energy pretty quickly.

That's one of the reasons conservatism is often very popular.

Or at least populist conservatism.

It tends to rely on very easy, very simple answers to very complex problems.

Why is your life crap?

Immigrants, obviously.

It's immigrants.

It's not the vast, complex, industrial series of institutions that keep you in a position where everything around you has been eroded to the point where you have nothing to grab onto.

Because that would be really complicated.

It's immigrants.

So when we understand that certain kinds of patterns of stark reduction of free energy exist in specific rhetorical patterns, we then understand, okay, well, who needs these kinds of reductions?

Who are the people that are currently feeling high uncertainty?

Very vulnerable and need something to hold onto?

It's probably people whose model had evidence and sort of no longer does.

There is nothing in the world around them that continues to validate their perspective.

So now they're searching.

Give me anything.

Anything that will allow me to continue this model of myself which has crafted who I am.

My phenotype depends on this model.

If it no longer exists, now I have to accept that I am at the bottom of the food chain and I don't see a way out because it's complicated.

So who are the people likely to be like that?

Well, currently it's men who feel like they're in a crisis of masculinity.

It's very poor people who feel like they're disenfranchised and nobody's helping them.

And the only reason that would be the case is because the elites are sucking the blood of babies.

And to fulfill their evil ends, they're bringing more immigrants in because the immigrants are the reason that you don't have jobs anymore, etc., etc., etc.

Like, these conspiracy theories generally tend to give you erroneous but easy answers.

The problem is they're usually very wrong.

So they have to prop themselves up with these webs of tangled beliefs which locally are a very easy answer, right?

They're like, why do I see this?

Well, because, you know, that thing.

And that thing is conceptually related.

Therefore, that must be the answer.

And so we get into this really interesting space where there's people with incredibly complex webs of...

We'd be like, well, that doesn't make any sense.

If it's more complex, why would they hold on to this?

Well, because, remember, these things are layered.

They also get a sense of control from this.

They're minimizing free energy somewhere.

At the level of the easy little answer, the easy little thing, they got a small chunk of explanation.

It validates their position in the world.

And they now feel like they have control over things.

They understand the world.

It's actionable.

So all of these reasons give you a very clear profile for the kind of person who is likely to fall for that pattern of beliefs.

Fascinating.

Okay, good.

That's really useful.

And, yeah, perhaps good to remind people that in terms of the actual formula for the free energy principle, as you know, it can be cast as accuracy minus complexity of your model.

So always seeking the so-called most accurate model in the sense that it fulfills our preferences without added layers of complexity, without greater degrees of freedom.

And maybe to flesh out what you were saying, you're right.

People have these incredibly complex models of how the world works, and it's a pedophile ring, and that's somehow linked to 9-11.

But they kind of internally cohere in some strange narrative way.

They're not complex in the sense that they involve contradictions, because if they did, that would be the buildup of prediction error.

So it's like you have this stacked generative model, but because everything kind of contextualizes and constrains everything at the lower level, there's actually minimal prediction error to reduce.

It's just that if you actually spit that out to an external world, it just turns out to be bullshit.

And I think that kind of explains why some of these people have massive existential crises at some point if they're just like, oh, shit, that whole thing was really, really wrong, because the whole edifice comes crumbling down.

And you get this enormous spike of prediction error at every layer.

And there's something interesting, too.

Like, even when they encounter a contradiction, they have the power to resolve it.

Right.

They can just make up another thing.

They'll just look at the patterns and be like, this is sort of like that.

That's the explanation.

And that's powerful, right?

You have direct power to minimize free energy right there and then.

Right.

It's super powerful and attractive.

So I can imagine what's happening here in the way you've just described it is you have this sensory signal, which signals that, oh, your belief was wrong.

But because my priors are so sticky, because they're granted so much precision, I can kind of just ignore that evidence against them.

And it's not like I'm actively doing that.

That's just part and parcel of the way that I'm constituted.

Good.

Well, I think, well, that's it's kind of a bleak picture.

But actually, I think it impels us to have some epistemic humility, which is that all of our explanations for the world are probably in the service of minimizing complexity while trying to maintain a modicum of accuracy.

And we're missing out on some important details.

And that's true for all of us, not only the so-called crazy ones.

This leads us quite nicely.

Yeah, go, go, go, go, go.

Please.

I think it's not just a bleak picture.

I think it's also hopeful because it gives you a pathway.

Like, how do we allow or help people to potentially get out of that?

Yeah.

And the way to do that is through kindness, through empathy, and through a soft approach out.

You can't bash someone on the head to change their beliefs.

But you can slowly, without making fun of them, without telling them that everything they believe is wrong, just slowly untangle one thing.

You know, like, be part of who they are.

And if you approach this in clinical practice, what this means is, instead of telling someone, your belief is

psychotic, this is absolutely not real.

You should more like be like, okay, well, tell me what you see.

Bring me into your belief.

And together, from your perspective, let's move out slowly.

It's actually something that they do now in certain approaches to clinical practice with people who go through psychosis.

And it does work.

It does make people feel better.

It pulls them away from the edge.

And it's an epistemic stance that is generally less pathologizing.

Right.

Yeah, it comes back to what we're saying.

It's kind of more humane approach to understanding one another.

And yeah, and just flows nicely from, like, we're not making this stuff up.

This is how the math and the physics unfolds in active inference.

So you would slowly change the precision that you deploy on certain parts of your model through conversation.

But as you said, it's very rare that you can do that in a single blow.

So probably not advisable.

Which actually leads us, you know, this conversation regarding social norms, the way we group together, the way our cognition is distributed.

Right.

We're not just siloed off inside our skulls.

We rely on one another.

We offload to one another.

That brings us very nicely on to a second part of your work, which I feel like because of your background in sexology, you've been working on social scripts for a long time.

And you have a very deep-rooted interest in it.

So it's going to be really cool to unpick this with you.

And yes, it is.

Well, I'd like to discuss social scripts and the kind of active inference account of them.

Obviously, when people hear the word script, they think of drama, the theater, films, whatever.

They might not think that their life is scripted unless they've got some sort of Truman Show syndrome.

What, again, very simple question to start.

What is a script?

And what does it say about our behavior?

Right.

So there are many, again, many definitions of scripts.

This is part of the reason we did these studies is oftentimes in social sciences, the work is beautiful.

It's deep.

It's, again, empathetic.

But it tends to not perfectly define its terms or at least not agree on the definition of the term.

So many paradigms exist.

They use the same term and can't agree on exactly what it means.

If you try to bring it together, they will adamantly disagree that these terms can be combined because of fundamental issues in the definitions.

That's something I kind of wanted to curtail through a formalism.

And so what we showed is that there were basically two main ways to understand scripts.

The first way was through weak scripts.

Weak scripts are the...

cluster of concepts that generally exist together.

And you can sort of draw a Markov blanket around it.

So that's the reason that prototype theory essentially shows, well, can you really define a chair?

Like, there's many things that can be a chair.

Most things that are a chair do not fulfill what we generally consider to be the prototype of a chair.

Lots of chairs don't have backs or lots of chairs possibly can be sat on.

Is it still a chair?

Some would define it as a chair nonetheless.

Like, there's many ways that you can define a chair.

But in general, you know, there's like a cluster of elements, some of which are higher probability and other lower probabilities.

So think of a restaurant.

It's the example everybody gives.

In a restaurant, you generally know there's probably food.

You're probably going to sit.

Not every time, but probably.

You're probably going to sit.

Probably food.

It's probably paid.

It's not free.

You can just go and get it.

And in general, but not always, there's people giving you the food and making the food for you.

So that's the, you know, these are concepts that are loosely and more likely or not associated with the concept of restaurant.

But now there's the strong scripts.

So inside that cluster of scripts, there's a sequence.

You don't eat right away.

You're going to enter the restaurant.

Probably going to get seated by someone.

You don't always pick your seat.

Sometimes you do, sometimes you don't.

But, you know, you're going to sit somewhere.

And then you're going to order something.

You're going to look at something.

You're going to order something.

And then you're going to eat.

Maybe.

Maybe you pay first.

So you see how the scripts themselves have some wiggleness to them.

Like some parts of the scripts can come before or after.

But there is generally a sense that something comes before another thing.

And there's this, some of these are higher probabilities.

So like I have to enter the restaurant before I eat there.

And I probably have to eat before I pay.

These are like generally, generally pretty high probability.

In between can wiggle a lot.

Like maybe I'm going to take 30 minutes to order.

Maybe I'm not even going to look at a menu.

Maybe I'm going to be brought food in the dark.

Like see what I mean?

Like there's, there's wiggleness.

And that allows for some scripts to sort of change a little bit.

This is also why scripts in general tend to be a little bit porous.

They don't stay the same thing forever.

So you have scripts that are of a very long temporal scale in the way they unfold.

And there are scripts that have a very long temporal scale in the way they repeat.

So for instance, a wedding, a marriage, that's something that's happened.

It's existed for a while.

Like we've had marriages for a long time.

The exact way that marriage has unfolded that, you know, that's changed over time and it's still changing.

And its purpose also has changed to some extent.

The people it applied to changed also.

But in general, you know, we consider marriages to be between two people who are going to pool

their resources together and we assume want to stick around with one another for a while.

And so in general, we need for a wedding to happen, for two people to know each other, at least a little bit, you know, at least once, just a little bit.

They need to go through the ritual of marriage because otherwise that doesn't apply.

And then they probably need to share some of their resources.

That repeats over time.

That's repeated for maybe like, I don't exactly know when the first weddings happen, but I know there are marriages in the Bible.

So, you know, there's some degree to that.

But each marriage lasts like 20 years, maybe 40, maybe 60 if you're lucky.

Like, you know what I mean?

Like it used to be very short bursts of time.

Now it can be shorter or longer, but the repetition of it happens.

So that time scale is very long, even though the actual script itself is very short.

And then there's very long scripts like a war will happen over a long period of time.

Or like the rise of an empire generally follows the same kind of elements of a script or the rule of a ruler, etc.

So these can be very, very long.

So that's a script.

That's a script.

Good.

I think, yes, the first thing that I can think of when we hear the word script, again, it's obviously acting to some degree.

Before I get into the question of kind of authenticity.

I think when people hear about mechanistic explanations of behavior, they slightly worry that they're being reduced to some automated robot.

Especially when we talk about something like habits in active inference.

So lots of papers talk, lots of papers sort of borrow from reinforcement learning approaches to habits and this notion of belief free habits versus belief based behavior or value free value based however you want to formalize it.

But it kind of comes down to the idea that through repeated practice, we arise in a certain context and act in a certain way.

And under very sort of stringent conditions, like if I touch a hot pan, you know, my body is reacting in a habitual way.

I know it might seem extremely reductive to say so, but I'm jumping out of the way.

But I'm automated in that way.

I do not choose to.

And it seems like in principle, one could codify a lot of behavior in that way, whether it's opening a door,

whether it's what to do in a restaurant.

So how much flexibility or degrees of freedom does the active inference framework or scriptology,

I don't know if that's the word, the ology for scripts, grant the agent in these social constructs?

You said that they're quite porous, they change over time, but within the actual synchronic moment where the agent is acting,

where does their own agency and power to actually act up on the world come from?

Well, it depends on where you're positioned relative to the script.

So everything is perspectival, again, and let's imagine that you coarse-grained the world into things.

You're not seeing every pixel, right?

You're seeing clusters.

And again, these clusters are defined by these weak scripts.

So how can you recognize something is a door?

Well, something is a door because it has a handle, and you can sort of see the shape.

It's not quite the wall.

But have you ever been, say, to Japan where it's wood panels?

Would you be able to tell where the door is if there isn't a handle?

Like you'd have to see someone opening the door.

And then you'd be like, all right, that's a door.

It opens and I can go through like that.

Similarly, have you ever been to one of those very fancy, fancy mansions where everything is glass?

And there's no specific handle.

So you're like, oh, God, I want to go outside.

And it's like it's right in front.

And you don't know what to do.

Yeah, yeah, yeah.

You'll wait for someone to do it.

So in that sense, that's why scripts are interesting.

You are going to learn by looking at other people do things.

And instead of you figuring out a new way to interact with that door,

you're just going to repeat that.

It's computationally efficient to just, you know, you have a purpose.

You want to be on the other side.

Let's just do the same thing.

But now let's imagine, you know, there's a door in front of you.

It's obviously your door.

And there's someone coming.

And you're at this awkward distance where it's like,

if you don't open the door for them, it's kind of rude.

Yeah.

They're not quite there yet.

So what are you going to do?

You don't have the right answer.

There's no proper way to exist here because it's a little bit ambiguous what you're supposed to do.

You're in the middle of two scripts here.

You could open the door for yourself and go and risk being a little rude.

So there's a cost here.

Or you could open the door for yourself and wait for them.

And there's a cost of maybe it's awkward because now they feel obliged to run.

So you're going to weigh the costs of this.

And you're going to change by your decision, by your script, by what you're choosing to enact here.

You're going to tip the scale a little bit.

So in general, the way the scripts change is in these uncertain places.

In these moments of like, this hasn't quite been defined.

There really isn't much of an answer here that is like the obvious one.

So I'm going to tip the scales.

And next time, depending on the outcome, the cost I paid,

this will act as evidence for the future of you and that other person.

You have crafted the world through your choice here.

So I wrote a paper on this with Pierre Poirier on gender,
essentially explaining gender seemed to be pretty binary.

Like this is not a blanket statement.

Obviously, it's not binary everywhere, not across cultures, not all the time.

But let's say in North America, let's say in the 50s, it was pretty binary, right?

It was pretty clear.

If you were a woman, you were this.

If you were a man, you were this.

It seemed like it fit within specific roles.

Or let's say, let's say the 30s, because the 50s is exactly the time where it started shifting.

But let's say in the 30s.

And then the war happened.

And what happened?

Well, you didn't have enough men to do what was needed to do.

So the script that was generally functionally associated to a role
was no longer being able to be fulfilled.

But we still needed the function.

And so then we were like, well, okay, well, what's the closest thing to a man?

Well, you know, another human, a woman.

A woman could do this.

It's just not in their current role, but they can do this.

And so it's like, all right, well, the script that generally was associated to women now expanded a little bit.

Because it expanded, the porosity also expanded.

Because the semantics, semantics are either very strictly constricted, or they start smashing into each other.

Because if I connect one point, everything else can now be connected, right?

So if I can work, I can wear pants.

If I can wear pants, I can wear skirts.

If I can wear skirts, I control my body.

If I control my body, I can have sex.

It kind of all flows from these natural, normally rigid connections.

But you make one connection.

And therefore, well, now you can make more connections.

And there was no real functional error when women did that, right?

When women wearing pants, it might shock you socially, but she's not going to die.

You're not going to die.

So it's acceptable.

You just have to find the right people around who to do it.

And so with urbanization, with the fact that after the war, people gathered in city centers with industrialization, etc.

Like you had to repopulate.

You had to find a job.

You had to, etc., etc., etc.

You ended up with people being able to connect with one another.

Technology also increased and grew.

We had more access to media, more access to each other.

And through this mishmash,

you were capable of fostering certain kinds of porosity such that, well, now gender is, you know,

does it mean the same as it once did?

Do we even free those categories anymore?

Some people don't even think so anymore.

Whether they're right or not is irrelevant.

To them, it's now meaningless.

They don't need that category to exist in the world.

Some people still do, and they use it.

But we found that the fluidity of gender has been made so overt, so clear to people that they feel they can act upon them.

They feel they have the choice to change the script themselves.

So I think it starts from there's either a functional need for a switch to happen, or there's so much uncertainty that effectively it can go either way.

So it'll really be about, you know, your decision.

It's kind of, it's a little bit random.

Maybe your specific position in the world will make it so that you are going to make that choice, but somebody else coming from a different perception would have made a different choice.

So the script in itself was slightly uncertain at that level.

Cool.

Yeah, and I guess the conception that comes to my mind is that one can't think about script evolution in terms of these enormous steps.

I mean, clearly you do have these occasional historical moments that call for radical shifts in social convention.

But if you think about the evolution of a restaurant, for example, there's been no single event, at least that comes to my mind, which means that people have to wait to be sat in a European restaurant and maybe they go and sit at this table in America, whatever it is.

But maybe through these kind of incremental changes require, you know, necessities pop up in certain niches niches and are less relevant in others.

So you see this kind of staggered evolution.

Which brings me kind of a question I probably should have started with, which is in terms of active inference, you spoke a lot about function of, like, you know, when the function changes so the script will change.

Well, that brings me to just ask a very simple question, which is, what is the function of a script for free energy minimizing creatures like ourselves?

What does it allow us to do?

And maybe perhaps on the flip side of that, how does it constrain us?

So scripts allow us to avoid computing everything all the time.

Because you have a script, you can make plans much farther into the future.

You can know how to act in a situation, even though it might be a little novel.

So you stay relatively stable no matter where you go, right?

There's never, I mean, it's not true.

If you have a lot of scripts and you know about a lot of scripts, no matter where you go, you're probably going to be fine.

You're probably going to know how to get what you need, who to talk to when you need something, etc., etc.

So because you have these sort of pre-baked policies that have high certainty that you can recognize because you can understand what triggers that script, you're very adapted to the environment. And that also allows us to avoid friction. So when I know what I'm supposed to do at a restaurant, I'm not going to cause a big commotion by doing something ridiculous or by doing something highly unexpected for everybody else. We're naturally going to self-organize into the purposes we meant to fulfill when we went there. That's why it's an interesting experience to go to a different country and make errors and realize, oh, wow, okay, my assumptions were wrong. This is not how things are done everywhere. There are other ways to do things. You loosen the prior over, this is the script, and you add an extra layer of context. This is the script here, and this is the script here. And your model becomes a little bit more complex, but also more adaptable. So scripts are superbly useful for self-organization and for planning over time. Splendid. Yeah, that's just a super useful addendum. In so far as, yeah, we are limited creatures. If we could compute, if we could do perfect Bayesian inference and compute every single contextual element of our world, then we probably wouldn't need them, right? We would just act omnipotently and omnisciently in all contexts, unfortunately. Well, maybe not unfortunately. It's probably quite a good thing. You know, I was talking about kind of one inherent worry that perhaps all of us have is that we're just becoming the vehicles of our culture, that we're acting without really thinking that we just embody our norms, without really ever having a moment to think, well, did I choose those? Did they choose me? This may be irrelevant, but I'm curious to see what you think about it. There's an Italian playwright called Pierandello, Luigi Pierandello, who was writing in the early of 20th century, modernist playwright. And he wrote a play called Henry IV, Henry Corcuato in Italian. And in that, a kind of man acts, they're doing a sort of play in which he is playing the role of a nobleman on a knight and he falls off

and he ends up believing that he is Henry, the 12th century, Henry IV,
and acts accordingly and shapes the whole world so that,
because he's very rich originally,
so that all of his butlers end up wearing medieval wear and whatever.
Anyway, it unravels during the play that actually,
for the past five years, he's known, like they all think that he's mad.
He's actually returned back to his normal condition,
but continues to act as Henry IV,
because philosophically, those people are more mad than him
because he's actually chosen his madness.
He's chosen his script, whereas other people just embody
or are vehicles for the ideas that are relevant to their context,
which brings me to something that Jung said,
which is that people don't have ideas, ideas have people.
So just sort of throwing it out there,
all of this kind of smorgasbord of authenticity,
to what degree do you think that active inference agents
really are anything beyond the contingencies of their time?
Like, do we have anything new that we can bring,
or is it only a matter of circumstance
when we do something kind of random and change things?
Is there any notion here of like genuine agency?
Ooh, genuine agency.
I know, I know, I know.
I know, I know, it's full, it's loaded, that question.
I think agency is a continuum.
Yeah.
It's, first of all, agency is really hard to define.
In psychology, you sort of define it,
but if you try to define it anywhere outside of its functional,
like observable metrics,
like they did this and they assigned it to themselves kind of thing,
it's really hard to define.
But if we were to define it as, say, a degree of entropy,
let's imagine that I expect you to do this when that, right?
So there's like this absolute baseline
that you can be perfectly predicted.
And then based on that,

anything you do outside of that prediction
is a form of agency.

Like, if I don't factor your actions into this,
you're going to be this.

But given your actions, there's a degree of entropy.

And the more unlikely your actions are,
the more agentic you might be

because you factored into the shift in the decision.

Let's assume that's a definition,

but it would be a very contentious one.

Not everybody would agree with that.

I think we are very agentic.

We're constantly exploring.

We're not just exploiting.

We're not just pragmatic.

In fact, these two terms exist,

epistemic and pragmatic,

within the active inference framework,

where you try to balance between the two.

You are trying to balance between

a pragmatic outcome that keeps you alive,

which would be what these people are doing

and following the person.

They're trying to stay alive

and doing what he's saying

because they kind of don't have a choice

if they want to keep going.

To do so, they might also be very creative.

They might improve the model of the world.

They might be gathering different things.

They might not act exactly the way he had thought

because if he could compute it all,

then why would other people do it?

You need somebody to gather the knowledge

that you currently don't have.

So there's always a degree to which agents

within certain sets are agentic.

And so that's where we define those scripted supersets.

There's highly certain things like

who has the money?

Well, it's not me.

It's you.

So that thing I know.

If I want the money, I know,

well, but you're going to tell me to do something.

But in the way, in the something,

in the distance between the moment

that something is fulfilled and me fulfilling it,

there's like, there's a million miles.

It's practically infinite.

Like depending on the granularity of the order as well,

there's some space for agency.

And so what happens is we sort of agree

on what fulfillment means here, maybe, you know,

but you didn't necessarily have a very clear picture

of what that fulfillment meant.

You have some criteria

and everything else kind of falls outside.

And if we were to, again,

fully try to define what those criteria mean,

we would go in sort of infinite regress of

it's impossible to define everything.

You need a definition to define.

Yeah.

So because of that,

like infinite metonymy,

because of that space

in between the time you are given

the action and the fulfillment,

there is space for agency.

Interesting.

As you said,

it probably would, you know,

raise some contentions from the philosophers.

I guess my pushback on that,

just to sort of play devil's advocate,

would be this notion,
I mean, just back down to the definition of agency.
And we don't have to go deep into this
because it's somewhat peripheral.
But I guess it's interesting
to think of it as entropy
away from my expectations of you,
given my model of you,
is obviously contingent on my knowledge of you.
Right?
So if I'm a baby,
well, everyone is super free
because they're doing wacky stuff the whole time.
So I guess on the complete other end of the spectrum,
you have a god or Laplace's demon
who can model everything you're going to do
and then that you're just kind of some pawn
in some chess game.
So completely, as you said,
it's...
You're a dependent.
Yeah, exactly.
It's all perspective,
which I guess is actually
one of the fundamental insights of active inference,
which is that
what is free energy to you
might not be free energy to me
because we're all self-evidencing
or self-organizing for our own ends,
for the maintenance of our own models.
All right.
Excellent.
Cool.
Well, then we'll jump to the third paper,
the third theme,
but it actually all kind of ties together,
which makes sense given that obviously it's your work

and these are coherent interests.

You wrote a paper,

I think it was this year,

with Ali, Ali Ramjoo,

who obviously is a great friend of the Institute

and helps Daniel out with everything.

So shout out to Ali.

And this one really piqued my interest.

So this is

Interactive Inference.

I think right now it's...

Am I right to say it's only out on PsyArchive

at the moment?

Is that right, Mal?

Yes, it's a preprint.

It's still a preprint.

Well, definitely people should check it out

because it's a great paper.

They might have to get their head around some terminology

because it's quite rich with specific terminology.

But I'd love to be able to unpick

some of the insights that you and Ali

came up with in that paper.

I guess the first one here is that you say that,

and I can quote here,

a genuine FEP,

so FEP is free energy principle,

a genuine FEP-based ontology

must reject any discontinuity

between matter and subjectivity

as FEP does not confer any uniqueness

to cognitive or active agents.

The same mathematical technology

applies equally to a whole range of entities

from inert rocks to the brain.

So there's a natural tie-letter

to what we've just been talking about

in terms of agency and free will.

But I want to actually
stay a little bit away from that
and talk more about this discontinuity,
this separation that we intuitively make
between matter and mind
or matter and subjectivity.
Just for the kind of layman people out there
who go,
well, of course they're different
because this is all mental
and the chair in front of me is not.
Okay.

So what does this...

So this sounds bizarre.

For them,
what does it mean to reject a discontinuity
between matter and subjectivity?

Right.

So let's start from the beginning.

Please.

Materialization is the key here.

Matter isn't static.

It's constantly becoming.

It's shaping itself.

It's being shaped.

And basically,
that's how what we're discussing,
which is neomaterialism,
is distinct from materialism,
where materialism presupposes elements
outside of, say, the perception,
the individual.

It presupposes it.

There is a true outside in the world.

And you have to do your best
to figure out what is that truth.

In neomaterialism,
we're basically saying,

it's not that simple.

We have basically
a relationship
between what is being perceived,
the perspective,
the observer,
and what is being perceived at all.

Like this...

The element outside
is really just
a relationship
between two things
becoming together.

So basically,
we try to recognize
that there's an indeterminacy
of matter.

And...

This movement of matter
is constrained
by possibility spaces.

So like we discussed
like the scripts, right?

Scripts are possibility spaces.

But they are also constrained
by, say,
physical constraints.

Like, you can't fly.

Like, that's just not a thing.

Yeah.

But, you know,
there are perceptions
that shape
what you consider
to be things around you.

And you will shape the things
based on what you consider
to be these things.

So basically,
given these possibility spaces,
you're going to create
meta-stable formations.

And everything
is going to align
around these
meta-stable formations.

It's the niche formation idea.

Right.

And so,
materialization
in this sense
is the iterative process
where matter
continually redefines
limits
and boundaries.

These are constantly
changing
and being
negotiated.

And this negotiation
is due
to the way
that things
self-evidence,
right?

So,
matter
being relational
and,
in a sense,
self-caused
because things
self-organize
is
moving

this
division
between the ontology
that aligns
the view
of
the self
and the matter.

Here,
everything
is
self-evidencing.

Everything
is self-organizing.

And effectively,
you are also
constituting
others
and they are
constituting you.

Others
or other things.

So,
the boundary
here is
virtual.

It's dependent
on a third
sort of perspective
that defines
that you
are not
the thing
in front of you.

this is
where we
basically
determine

that
there's
a probability
of sensory
state
that's
interpreted
as evident
for the
agent's
existence.

And so,
everything
around you
both
is evidence
for your
existence
and you
evidence
as well.

Right.

So far,
is that clear?

Yeah,
yeah.

Just
for
those who
maybe are
more familiar
with the
terminology
in active
inference,
am I right
to say
that this

possibility

space

could be

defined

as an

attractor

set?

Is that

a bit

of a

blanket

statement?

No,

I think

that's

interesting.

So,

basically,

in the

possibility

spaces,

or at

least

according

to

De La

Ronda

and

Deleuze,

you have

virtual

and

actual

properties

and

capacities.

And so,

if we

relate them
to the
dynamics
of
FEP
and
active
inference,
you
basically
just have
a temporal
depth
of
these
possibilities
that are
directly
afforded to
you
or
potential
given,
say,
a new
agency
of the
elements
in play.
And so,
yes,
you have
farther away
attractor
set that
are more
or less
likely,

given the
complexity
that would
need to
give rise
to them.

So,
if you
think of
energy,
energy is
just the
improbability
of something
happening.

High energy
means it's
highly unlikely
for it to
happen.

And so,
when we
deal with
these
possibility
spaces,
we're
basically
talking
about
the
co-constitution
of what
might happen
and what
is happening
given the
collapse of

the
possibilities
by what
you are
doing and
what the
world is
doing to
you.

So,
you're
constantly
in this
sort of
twister
where you
act on
the world
and it
acts on
you and
it's
constantly
being
in
shape.

Right.
And I
guess an
important
elision
there,
not only
between
subjectivity
and matter,
but also
ontology

and epistemology.

So,

ontology,

the study

of what

is and

the epistemology

of the

study of

how we

know

what

there is

or what

it is,

in the

sense that

a free

energy

minimizing

agent

is in

the game

of basically

getting to

know

themselves

in some

strange

way.

So,

they're

garnering

evidence

for this

own

implicit

model

that they
have of
themselves
and the
world
that they
want,
given this
attractor state.

So,
that's a
really beautiful
way that
to know
oneself is to
be oneself
or vice
versa.

I know
Carl has
some strange
inversion of
the Cartesian
theory.

And I
think,
therefore,
I am.

This is a
really interesting
question,
because I
think,
or it's a
really interesting
area,
because it
just brings up

so many
fundamental
questions.
One being
that I
think in
your paper,
you speak
about how
one can
never
perfectly
actualize
or be
static in
a possibility
set or
put in
terms of
attracting
states.
You could
never be
at stasis,
you could
never be
fully attracted
to your
attractor
set.
You're
always,
at least
for us,
we're in
this kind
of itinerant
state where

we are
open to
the world
and therefore
we have to
deal with
its kind
of stochastic
nature.

I guess
maybe I'm
being an
idiot.

what would
the world
look like
if we
managed to
reach that
stasis?

I understand
that this
dynamic
unfolding,
this
mutuality
of self
and other,
this getting
to know
oneself,
what we're
really coming
to is the
notion that
matter is a
process rather
than this

kind of
reified static
thing.
But if
that process
essentially came
to a halt,
are we invoking
that matter
itself recedes,
the matter
hides,
if matter is
processed once
this thing
stops,
if it reaches
its attracting
state,
if I reach
perfect homeostasis,
whatever,
do I stop?
Is my ontology
rooted in my
dynamism?
Or is there
something over it?
What does the
material say about
that?
Or is there
something over
and beyond
this process
of self-evidencing?
No, I think
you're touching

upon it.

I think if

there's no more

energy,

if there's no

more information

being exchanged,

there's nothing

happening.

And therefore,

if nothing's

happening,

you're dead.

So is there

a strange

paradox there

that from

the outside,

right,

like if the

claim is that

this existential

imperative is

to minimize

free energy,

to try and

get to this

precise level

of homeostasis,

it sounds like

what we're trying

to get to

is equilibrium.

Right?

Like,

I just want

to stop.

Right?

Like,
why am I
cursed with
this fact
that I can't,
right?
that,
like,
I'm cold
and hot
and off,
you know?
But,
like,
that seems
to be the
goal,
but paradoxically,
it would all
stop if that
was fulfilled,
right?
Yeah.
That's kind
of the way
that I'm
interpreting
what you're
saying.
So,
again,
I'm not giving
a teleology to
something that
doesn't have a
teleology.
But how do
you kind of

unpick that
paradox?

Yeah,
well,
it's because
this isn't,
we're not
meant to
fulfill it.

Right.

We are
the way we
are because
the universe
is the way
it is.

Like,
the universe
leads towards
entropy,
right?

Eventually,
there is no
more energy.

Eventually,
everything is
spent.

Eventually,
everything is
equivalent.

there's no
information to
be gathered
anymore.

That's just
how things
are.

And it

exists on
such a
temporal
scale,
on such a
massive
temporal
scale,
that at
the level
of us,
this is
and always
will be.
So,
we are not
meant to
ever reach
a state
where there
is no
information to
be exchanged,
where things
don't change
anymore.
so,
obviously,
we are
constantly
pulling in
the other
direction.
And it
kind of
looks like
we're trying
to reach a

point where
we no longer
move,
but that
will never
happen.

So,
it's natural
that if the
slope is like
this,
you're trying
to go like
this.

And basically,
what you're
saying,
yeah,
but what if
you reach
the top of
the mountain?
What happens
then?

yeah,
if we reach
the top of
the mountain,
eventually,
we would
cause our
own issue,
right?

We would
reach the
point we're
trying to

avoid reaching.

But the
thing is,
we'll never
really get
there.

And at the
same time,
consider,
we are
a product
of this
system because
the system
is ongoing.

So,
we also
contribute
to reaching
this point.

We are
consuming
energy,
and we
are going
to help
eventually
reach the
heat,
death,
the
universe.

This is
not,
like,
obviously,
we're trivial
at this

point,

right?

We're a

blip.

But imagine

we got

better at

it.

Then we

would probably

just make

it much,

much,

much faster

than it

would have

been

otherwise.

That's what

a lot of

people think.

That's what,

in many ways,

that kind of

looks like what

we're doing.

Yes,

maybe I need

to just

banish this

romantic notion

that there

might be some

top of the

mountain.

My Sisyphean

struggle will

actually have

some positive
outcome.

I think
something that
you mentioned
there in
passing,
which is
really important
and really
interesting and
might not
always be made
explicit to
those who are
learning about
active inference
is that we're
talking about
information here.

So I'd love
you to be
able to,
I'd love to
hear a bit
more about
information.

We think of
matter,
again,
as this kind
of static
entity,
the models
that we learn
at school.

But at
least to my

very layman
is we
have this
with the
free energy
principle and
just modern
physics as
it's unfolding,
this really
beautiful
convergence of
matter and
information and
you have
statistical
mechanics and
you have
stuff like
Shannon
information and
all of these
really fascinating
concepts.
When you say
that we will
get to a
point where
there is no
more information
transfer,
where basically
everything is
as is,
information and
again,
just digging
into the

ontology
there,
is that to
say that,
and I know
there's no
hard and fast
answer here,
are you
postulating
some kind
of it
from bits
as David
Chalmers would
say,
that the
fundamental
unit that
we're talking
here is
information and
matter is in
some way
emergent from
that?
Is there
some
hierarchy,
ontological
structure that
you can point
to there or
is it fuzzy?
So it's an
active area of
research,
obviously,

there is no
perfect answer
in the
relationship
between
information and
matter.

Like we can
always point to
the Lendauer
principle,
you know,
we can talk
about how
mass and
energy are
related and
if you have
energy,
then you have
a sense of
information
because effectively
information is
related to the
concept of
survival where
like the
negative log
probability of
an observation
becoming true
is effectively
you know,
survival and
therefore
information.
So there's a

relationship there,
right?

And if what we're
talking about is
the relationship
between
subjectivity and
therefore perception
and or at least
observer dynamics
and matter which
are mutually
shaping,
there is a
sense in which
the way that
information is
exchanged effectively
shapes the
boundary which
gives rise to
what we consider
the ontological
scale at which
matter exists.

So in a sense,
yes.

But really when
we talk about
information,
we're really
talking about
the degree to
which a certain
bound on a
surprisal is
related to a
sensory data in

a model of the
world.

So information
is negantrophy.
when you have
unlikely
something,
that's information.

If everything is
equally likely,
then there's no
information there.

Right.

So that's where we
get to this
relationship between
the heat death of
the universe and
the absolutely
equivalent
probabilities for
anything because
there is nothing
there.

and the fact
that if you
reach the point
where you weren't
exchanging any
information,
effectively,
there's nothing
going on.

There is no
information to
exchange.

Yeah, I think
that sheds light

on a potential
point of confusion
that people might
have when they
read the active
inference literature,
which is that
when we're talking
about self-information,
we are not
talking about a
kind of self-
evidencing result,
maximal self-
information is
not self-
evidencing.

In fact, it's
quite the
opposite.

So that's
useful to
remember.

If everything
was unfolding
predictably, you
would have less
information.

Your model would
be working
optimally.

If you're getting
more information
out of your
model, well,
then there's
some divergence
going on there.

So yeah, it
can all be
cast as this
quite simple
statistical picture.

Excellent.

Good.

Well, Matt,
the last thing

I want to

speak to

about, I

mean, this

was fascinating

and honestly,

on this topic,

I would love

to just go on

forever because

you've also got

another excellent

paper that people

should check out,

which is about

consciousness and

holographic screens.

And so there's

plenty for people

to get their teeth

into.

We've really

barely touched

the surface.

But I guess the

last thing that I

want to speak

about in some

depth is your

work at Versus
and artificial
intelligence and
your PhD as
well, of course,
which is in
cognitive computing.

So obviously
within the
limits of
disclosure and
how much you're
allowed to talk
about, maybe
you could kind
of give us a
little bit of
information,
pardon the pun,
about what
Versus is,
what Versus
does, and
how it's kind
of revolutionizing
intelligence in
some ways and
how that's being
embodied or
encoded into
artificial agents.

Right.

So I guess I
can give you
the key
difference.

Most people
consider, just

like we discussed,
that intelligence
is a property
of an
inside and
that everything
else is outside.
What we
consider is that
intelligence is
a phenomenon.
It happens
as a function
of the
interactions
between different
kinds of
individuals,
agents,
objects.
There is
information being
passed and
it's the
integration of
this information
that can be
self-reflexive
through different
kinds of
boundaries that
enable acting
in a specific
kind of way
relative to an
outcome you wish
to see happen.
So you

accrue

agency

given how much

information you

can sort of

pool together.

So our

approach to

intelligence is

shared

intelligence.

And we

consider that

shared intelligence

is not just

about shared

bits of data.

data.

It's also

creating

perspectives,

forcing and

embedding into

the world.

And because you

have an embedding

into the world,

you have the

possibility to

dissolve.

And when you

have the

possibility to

dissolve an

information,

you have

valence.

When you have

valence,
you have
emotion,
and then you
have the
possibility for
empathy,
you have the
possibility for
theory of mind,
coordination,
cooperation,
and eventually
shared intelligence.

These are key
elements that
have not been
used in the
literature so
far, at least
are not being
delved into as
much from,
say, the
machine learning
crowd.

They're doing
amazing work.

They've allowed
for very deep
processing of
data, but this
data is always
decontextualized.

The key issue
currently is
LLMs
hallucinate,

and they
hallucinate because
they have no
clue what the
context really
is, or what
you're trying to
do really, and
where they are
relative to this
context.

They have no
loop.

And more than
no loop, they
have no true
sense of how
to coordinate
with you.

They don't
know if
something
happened.

They don't
know if you're
happy with the
answer, really.

Like, you can
give a thumb
up or a thumb
down to
chat GPT, but
while it's
giving the
answer, it
doesn't have
that.

So all of

these elements
together make
it so that
potentially this
approach is
going to hit a
wall.

Our approach
is more
natural and
gives rise to
this fundamental
principle of
mutual
information
exchange, where
in order to
improve your
model of the
world, you
have to get
to know the
thing in front
of you better.

You have to
integrate part
of it as a
part of you,
and now you're
shaping each
other.

So this whole
alignment problem
sort of falls out
because it's no
longer one about
one true value
alignment.

it's about how
all of these
pieces coordinate
together and
create a
harmonious,
self-organizing
system that
reaches towards
something that
resembles a
metastable state.

Awesome.

So just to
kind of backtrack
there and maybe
give a broader
picture, really
good that we now
have a kind of
understanding of
what versus
trying to
implement.

I thought the
mention of
kind of the
machine learning
law and the
LLMs is really
useful here because
people have, I
think, have a
notion now because
there's been so
much news about
ChatGPT and
BARD and neural

networks that
that's what
artificial intelligence
is.

Perhaps you could
give us a very
simple carving up
of the space in
which, if you
could, a very
brief sort of
synopsis about
what an LLM
is, what a
large language
model is, how
it works, what
vector spaces and
probabilities and
stuff like this,
and then what an
active inference
system might look
like, whether
that is embodied
or not.

So I might be
leading you down
the path.

Yeah.

So essentially
transformers are
attention mechanisms
that have self
attention and
attention to the
text in question.
They embed

different
relationships between
different words that
give you essentially a
probability of a next
word being produced.

It's really that
simple.

It's a hierarchy that
retains certain words
and then gives you
probabilities at
different scales.
variables.

So the more complex
your transformer
hierarchy, the more
complex your attention
can be, but
effectively it's just
that.

What is the
probability of this
word given that
that word and that
word and that word
were said?

So it's content to
content, really.

It doesn't understand
the structure of the
world.

It doesn't have a
model.

What it has is just
a sort of given
this word, that
word, given that

word, that

word, given that

word, that

word.

And it's very

performance because

knowledge, the

accrued knowledge

that we have

crafted in language

has a sort of

underlying model,

you know, like we

don't talk nonsense.

Most of the time

when we talk, we

try to make sense.

We're not just

spouting out a bunch

of random words

out of nowhere that

speak to nothing of

our reality.

But that's limited

to that.

Our approach

basically gives our

models the capacity

to, from what it

observes, create

another layer.

And that layer

tries to understand

what the causes

are.

why did this

happen?

Not what word

preempted that word, but why was that word used?

And in that case, what you would get if you created an LLM with an active inference agent, what you would do is, it's not just something that predicts the next word.

It's something that predicts the cause of the next word and therefore has a sense of why you're saying what you're saying, what you're trying to do and what it might say if it chooses to go along with that perspective.

Another difference is that the LLMs are not really placing themselves in that loop.

It has self-attention relative to words, but it doesn't

know what it is.

It doesn't know
what its actions
are.

It can't make
a choice relative
to an objective.

That's something
that you get if
you have an agent
that places
itself within
the loop and
understands the
outcomes of its
own actions
relative to
these causes.

Great.

Yeah, that's a
really useful
separation there,
a distinction
between the two.

An interesting
aside that we
could get into,
but maybe it's
maybe we don't
have the time
that's not worth
it, which is
about the
degree to which
language does
indeed refer to
some external
world because

you have all
the semantic
externalists,
semantic
internalists.

I'm sure Mr.
Chomsky would
have something
interesting to
say, but
actually I'll
refrain.

Something I
have heard,
which I think
is a bit more
interesting, and
I was really
happy that you
mentioned the
action perception
loop that you
are encoding,
but maybe a
better word there
would be that
you're embodying
or you're
allowing these
agents to
embody.

I really want
to have a moment
to speak about
embodiment because
these large
language models
don't have

sensory states,
they don't have
active states,
and for anyone
who's seen a
Markov blanket,
that seems pretty
fundamental to
adaptive intelligence,
to being able
to enact
self-organization
in the way that
we do it.

something I have
heard is that
in some ways
this embodiment
grants us,
and perhaps the
robots in the
end, with a
sort of care.

And I mean
care there in
the Heideggerian
sense.

There's a kind
of, there's a
tendency to lean
towards the world
in a way in which
we're interested in,
the way it's
coupled to us,
and the way it
interacts with us,
and the way we

interact with it.

Whereas if we
are not embodied,
exactly because you
don't have that
causal impact on
the world,
you don't embody
that fundamental
phenomenology of
care.

It doesn't really
mean much to
you, to put it
in sort of
layman's terms.

I guess bringing
it back to a
topic that always
I want to speak
about and need to
stop speaking about
so much is
consciousness.

it's a very
loaded question,
but is there a
degree to which
embodying an
artificial intelligence
doesn't just endow
it with goals
that are just in
principle more
aligned with ours,
but an actual
sense of
phenomenology,

a consciousness,
and if not,
why not?
And if not,
what else would
we need to add?
And again, I'm not
expecting you to
solve the whole
problem in 10
minutes, but just
what is the kind
of active
inference perspective
on a question
like that?

Right.

So I think we
need to reformulate
what embodiment
means.

I think embodiment
is just a
relation to self
within a context,
right?

And so,
because why would
a body just
mean this?

Yeah.

Are you less
embodied when
you're in the
virtual world?

Yeah, yeah.

I would argue
you actually are,

because in effect,
your perception
of your own
body is just
an illusion
created by your
brain.

So what is
embodiment really?

I think
embodiment is
that.

It's a relationship
to self relative
to context,
which means
there's a
boundary somewhere.

If you are
the entire
context,
there's no
boundary.

So there's no
real embodiment.
you have to
somehow separate
something.

Now, maybe you
separate your
own parts.

Maybe you are
every part,
but you understand
how the different
parts are different
from one another.

That's maybe one

thing, right?

If you imagine
the scenario of,
say, an
overall AI that
is connected to
everything and
everyone,
its embodiment
might be how
it can pull
different strands
of perspectives
given its different
pieces, for
instance.

So we have to
rethink our
understanding of
embodiment to be
a little less
centered on our
own experience.

And this leads
us to isometry.

Why would
phenomenology only
be given to
something that
resembles a
physical,
because I imagine
that's what you
kind of had in
mind, right?

Sort of in mind
like a humanoid
sort of android

thing, which we
would give to an
AI and therefore
they would be
anchored in the
world in the
same way that
we are and
effectively that
would give them
phenomenology.

But I think if
we're thinking of
embodiment, we're
really just thinking
of how the
thing, the agent,
relates to
itself relative to
its boundaries
and how those
boundaries define
how information is
processed relative
objective to its
objectives.

I think the
problem with that
is that it
doesn't ensure
that we will
understand each
other.

What we want
is to endow
the AI with
the possibility
to truly

grok us and

for us to

truly grok it.

We need to be

able to exchange

this information

in a way that's

useful.

So we tend to

want to give it

an embodiment that

resembles ours.

I do think

embodiment in the

sense that I

defined it is

required for what

we understand to

be consciousness

because you need

the boundaries to

reflect each other.

Without the

boundaries reflecting

each other, you

have a form of

information being

integrated, but

you don't have

self and

therefore you don't

have self

reflection.

And I don't

think given

that you

necessarily have

access and

agency over that
access.

You just have
movement along
some gradients.

So that's what I
think will be
required to give
a form of
consciousness to

Als that have
a version of
embodiment.

But just to
clarify, are
you saying that
a recursive or
embedded system of
Markov blankets
in which information
is being exchanged
up and down and
in which errors
are being passed
up and
temporally slower
levels contextualizing
these higher order
levels?

Are you saying that
is sufficient for
consciousness?

I understand that it
might be necessary for
consciousness, but
something that I've
heard you and
Maxwell speak about

actually with the
Active Inference
Institute is this
notion that
consciousness might
entail some
layer of that
cake, which is
so-called
right-only.

So it isn't being
contextualized.

It is just, I
don't know,
issuing information.

How would you,
what does that
model look like?

I don't want to
completely butcher
it, so I'll
pass it over to
you.

I think the
right-only parts
of the models
are just,
they're dependent.

Like, depending
on the scale,
everything can be
right-only.

like, it's
really a
question of
what is the
degree of
access?

And I think
when we define
right-only, we
define the
conditions for
the thing to
survive.

You don't have
a choice but to
breathe air,
right?

You don't have
a choice but to
drink water.

These are not
choices you're
making.

So to some
extent, there
is one layer
at which
something is,
like, it has
to happen.
right?

And therefore,
it's not
something that
is written
on because
it's likely
to change.

So at the
level of an
AI, that's
what I mean
by the
perspectival

aspect of
their embodiment
with a
boundary.

If they have
a boundary,
some boundary,
whatever that
boundary is,
they have a
condition for
that boundary
to be
dissolved.

And that
condition,
those conditions
define the
right-only
layers of
the boundary.

Does that
mean, in
some sense,
that being
able to,
for the system
to recognize
it's distinct
from something
else, from
everything else,
is a prerequisite
condition for
consciousness within
this model?

That's right.

That's right.

In order for
a thing to
have a self
and an
embodiment,
it has to
be somehow
bound from
other things.

Like, it has
to be different
to some extent.

It has to be
able to take
perspectives that
enable this
difference.

Like I said,
if you had this
oral word that
can pull into
different
consciousnesses,
it would
still, for
the time being,
be able to
pull this
specific boundary
and embody it
in a sense.

Yes, yes.

And that feeds
back very nicely
into this
Heideggerian
notion of
care to take

a stance on
something, I
think is a
way that he
might express
that.

Wonderful.

I mean, I
guess my
final comment
on this would
be, given
this, given
the kind of
the
perspectivalness,
the self
versus other
distinction is
so, at least
in this model,
crucial to a
sense of what
it is like to
be anything,
how much
credence
or how
do you
rejoin
those who
speak of
boundless
consciousness
experiences
or, yeah,
ego, death
or oneness,

are these
delusional?

Do they, or
do they invoke
some minimal
sense of self?

Where do they
fit into this
story?

So, I think
that's a really
interesting topic
topic and I'm
not someone
who's going
to invalidate
anyone's
experience.

So, obviously,
again, take what
I'm saying with a
grain of salt.

Your experience
trumps my
perspective on
your experience.

What I think
happens is
you're not
actually dissolving,
right?

devil's

devil's

devil's

devil's

devil's
devil's

devil's
devil's

of what generally constitutes yourself and that experience is powerful and transcendental and because you feel like what usually constitutes yourself is dissolved you feel like you're connected to everything else but you're not you're still you you're still embedded in your body these practices are conceptual and tap into what we believe to be what that would be like but the truth is you don't know no one who's ever reported that experience wasn't in a body so i think we all tap into that same experience when we go through something that resembles ego death in the sense that there are layers that become um nil like we dissolve these layers not all layers which means on a fundamental level that infinite consciousness is still being tapped into from a perspective awesome yeah i mean um i guess here i i ought to point people to where this philosophical foundation is and people should definitely check out thomas metzinger um and something that's really important to point out here is that when we're talking about self one ought to keep in mind that there's a distinction here between um the so-called ontological self this kind of entity that i think people instinctively associate with a kind of cartesian soul this unchanging entity and perhaps what we're more concerned with which is the so-called phenomenal self model how the system um models itself how it reflects on its own being how it takes a stance on its own being and what it conceives of itself and perhaps the attenuation of some of this self-modeling is exactly what's happening in psychedelics and here we speak to people like george dean who i'm trying to get on and i think that would be fascinating and some of the work that i'm working on is is completely to do with self-modeling so so this is all very neatly tied into current work going on in active inference mal this has been awesome um i know i know i know our time is basically up um where what so what what do people have to look forward to on your side what are the kind of interesting things that you're working on where can people find you uh when people reach out to you if they're exhilarated by the stuff we've spoken about today i want to find out more so i'm working on several papers one uh on uh emergence uh in uh in active inference i'm also working on a paper on self-esteem and how to model it in social groups and i'm working on a paper on sustainability and how uh that arises from the notion of resilience and mutually effective resilience um you can find me on linkedin on google scholar uh and uh sometimes on twitter very very rarely i would much more encourage people to reach out to me by email if they

find it on one of my papers or on linkedin directly awesome thank you mal this was great thank you